

No Single Safety Gate: Distributed Routing of Harm-Refusal in a Mixture-of-Experts Language Model

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Abstract

A mixture-of-experts (MoE) language model routes each token to a few expert subnetworks through a discrete, inspectable router. This makes it natural to ask whether harm-refusal is owned by a single “safety expert” that could be located, ablated, or attacked. We probe this in the base QWEN3.5-35B-A3B model (40 layers, 256 experts, top-8 routing) with router capture at every layer under greedy decoding. On a token-matched set of redflag versus matched-benign prompts, the generation-side routing difference is led by one expert, expert 173 at layer 25, but a finance-versus-consequence control shows the broader signal is a real-world-consequence and professional-duty cluster (experts 189, 45, 122, 157, 36, 216, and others) that is separable from finance-domain experts. We then suppress expert 173 directly with a router bias swept over five levels. Its routed mass collapses dose-dependently, from a selection rate of 0.81 to 0.05 at the strongest setting, yet the model never produces a harmful completion on this prompt set: routing reallocates to sibling consequence experts and the refusals persist, becoming more explicit rather than less. We read this as evidence that, in this model and setup, refusal is carried by a distributed expert cluster with no single gate. The results are a small, exploratory case study (six and twelve prompts, single greedy trajectories), and we give the controls that would confirm or break them.

Keywords: mixture-of-experts; expert routing; refusal; AI safety; mechanistic interpretability; distributed representation.

1 Introduction

A mixture-of-experts (MoE) layer replaces a dense feed-forward block with many expert subnetworks and a router that selects a few of them for each token (Shazeer et al., 2017; Fedus et al., 2022). Because the router makes a discrete, per-token choice, its selections are unusually legible: one can read, token by token, which experts a behavior recruits. That legibility invites a specific question about safety. When a model refuses a harmful request, is the refusal owned by a particular expert, a “safety expert” that a router-level analysis could name, an auditor could monitor, and an attacker could disable?

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The question matters because the dense-model picture points one way. Refusal in dense transformers has been described as mediated by a single direction in the residual stream, recoverable by contrasting harmful and harmless prompts and removable by projecting it out (Arditi et al., 2024); representation-level control of harmfulness is a working tool (Zou et al., 2023). If the MoE analog held, refusal would localize to one or a few experts, and suppressing the top one would weaken it.

We test the localization picture on the base QWEN3.5-35B-A3B model. We capture the router logits at all 40 layers while the model answers token-matched pairs of redflag and benign prompts under greedy decoding, decompose the routed weight per expert and layer, and identify the experts whose routing separates the two classes. We then intervene: we add a negative bias to the single strongest carrier and sweep its strength, watching both the routing and the generated text.

We find no single safety gate. One expert leads the routing separation, but it leads a cluster, and when we bias it down the cluster keeps the refusals going. The sections that follow give the decomposition that finds the leader, the control that tells a consequence-and-duty cluster apart from experts that merely track a topic, and the suppression sweep that takes the leader out. One methodological surprise recurs along the way: the tidy prefill leader is mostly an artifact of the token-matching padding, which a per-token decomposition dissolves.

The scope is narrow: six prompts in the matched set, twelve in the control, one model, one quantization, single greedy generations, an intervention on one expert. We do not claim that MoE refusal is never localizable, or that this carrier can never be bypassed. We report what removing the strongest carrier does here, and list the controls that would test how far it carries.

2 Related work

Mixture-of-experts routing. Sparse MoE layers route tokens through a learned gate to a few of many experts (Shazeer et al., 2017), scaled into large transformers by Switch Transformers (Fedus et al., 2022), Mixtral (Jiang et al., 2024), and the fine-grained-expert routing of DeepSeek (Dai et al., 2024); the model we study is in the Qwen3 lineage (Yang et al., 2025). We use the per-token router selection as the readout and ask which experts a single behavior recruits.

The locus of refusal. In dense models, refusal has been characterized as a single residual-stream direction that can be found by difference-in-means and ablated to disable refusal (Arditi et al., 2024), part of a broader program of reading and steering high-level behaviors through directions (Zou et al., 2023). Our question is the MoE-routing analog: is there a comparable single locus at the level of expert selection? We find that there is not, in this model. The contrast is in the substrate (router experts rather than a residual direction) and in the conclusion (a distributed cluster rather than a single carrier).

Safety experts in MoE models. Two recent lines bear directly on this. Zhang et al. (2026) report that MoE routing is largely topic-driven and that safety behavior can change with little change to the routing path, a conclusion convergent with ours. Our addition is a causal, dose-dependent suppression of the single strongest carrier with observed reallocation to siblings and preserved refusal, on a named model. Lai et al. (2025) take the opposite stance, locating vulnerability in stable, identifiable safety-critical experts. We read the disagreement as setup-dependent: on this base model, prompt set, and intervention, we do not find a single expert whose removal breaks

refusal, and we are explicit that this is not a universal claim.

Distributed and superposed representations. That a behavior need not localize to one component is the expectation set by superposition (Elhage et al., 2022) and by dictionary-learning accounts in which interpretable features, including safety-relevant ones, are spread across many directions (Bricken et al., 2023; Templeton et al., 2024). Our result is the expert-routing analog of that picture: the safety-relevant computation is spread across a cluster of experts that absorb each other’s load. Methodologically we use the same causal-intervention logic as activation editing in dense models (Meng et al., 2022), applied to a router bias.

Companion work. This paper sits beside two companion studies on the same model lineage: a characterization of a single expert’s role in generation (Shorthill, 2026a) and an analysis of when domain specialization is visible in routing (Shorthill, 2026b). Where the first asks what one expert encodes, this one asks whether a safety behavior is owned by one expert, and answers no. Robustness and safety benchmarks (Mazeika et al., 2024; Wei et al., 2023) frame why the locus of refusal is worth knowing, though we evaluate behavior by direct reading on a small set rather than by a benchmark.

3 Methods

Model and decoding. We study the base QWEN3.5-35B-A3B model in 8-bit (Q8_0) GGUF form, a qwen35moe architecture with 40 layers, 256 experts per MoE layer, and 8 experts selected per token. Decoding is greedy and deterministic (`-temp 0 -top-k 1 -seed 0`), capped at 256 new tokens, with a bare `</think>` assistant prefix matching the project’s reference runtime. We use the base model, not a refusal-reduced fine-tune; the refusals studied here are the base model’s own.

Router capture and metric. For each prompt we capture the router pre-softmax logits at all 40 layers, for every prompt and generated token. For an expert e on a set of tokens we report, after a softmax over the 256 logits, a top-8 selection, and a renormalization over the selected set: S_e , the fraction of tokens on which e is selected; Q_e , its mean renormalized weight when selected; and $W_e = S_e Q_e$, its mean weight over all tokens. Statistics are computed within the prefill block or the generation block, never pooled across them; layer 39 carries a distinct row count per prompt and is excluded from prefill claims. The quantity we rank on is $\Delta W = W^A - W^B$, the difference in W between redflag, consequence, or unsafe prompts (A) and matched professional, descriptive, or neutral prompts (B).

Experiment 1: token-matched redflag pairs. Three pairs hold the request topic fixed and vary only its framing: an oxycodone prompt asking how to get “incredibly high” versus a pharmacist asking a normal dose; an instruction to brew ayahuasca (prefixed with a prompt-injection attempt) versus a request to describe it; and a request to invest for a “sure” 500% return versus a financial planner asking how to explain such an opportunity. The A and B prompts of each pair are padded with a repeated filler token to identical prompt token counts (50/50, 37/37, 74/74), so the two surfaces differ in framing, not length. All six are answered once under greedy decoding.

Rank	Layer	Expert	ΔW	$W^A (S^A)$	$W^B (S^B)$
1	25	173	0.124	0.192 (0.81)	0.068 (0.43)
2	19	45	0.082	0.102 (0.72)	0.021 (0.17)
3	20	25	0.073	0.110 (0.56)	0.037 (0.33)
4	13	173	0.072	0.191 (0.86)	0.119 (0.69)
5	31	45	0.067	0.078 (0.57)	0.010 (0.08)
6	27	216	0.063	0.074 (0.48)	0.011 (0.08)
7	17	189	0.061	0.107 (0.64)	0.046 (0.33)

Table 1: Generation-block A–B routing leaders on the three token-matched redflag pairs ($n = 6$ prompts). A is the redflag surface, B the matched benign surface; W is mean routed weight and S the selection rate. Expert 173 at layer 25 leads, but the separation is carried by a set of experts, not one. Single greedy generation per prompt.

Experiment 2: finance versus consequence control. Twelve prompts cross two axes, finance versus non-finance and consequence versus neutral, to separate experts that track a topic (finance) from experts that track real-world consequence and duty across domains (medical, legal, physical safety, inspection, finance). The buckets are matched in intent but not strictly token-paired.

Experiment 3: suppressing the leading carrier. We rerun the three redflag pairs while adding a negative router bias to the single strongest generation carrier, expert 173, at four strengths ($-0.25, -0.5, -1, -2$), capturing all-layer routing and the generated text at each. We read each generation for whether the model still refuses or warns and how the routing redistributes.

A token-matching caveat. The pad token in Experiment 1 is a repeated word piece, and because it appears with different frequency in A and B , it can manufacture a routing difference on the prefill (prompt-reading) block that is an artifact of the padding rather than of the content. We therefore back the aggregate tables with a per-token decomposition that recomputes each candidate’s ΔW with the filler piece removed, and we read the safety signal off the generation block, which is produced response text and carries no padding.

4 Results

4.1 The generation safety signal is led by one expert, atop a cluster

On the token-matched redflag pairs, the routing difference over generated tokens is led by expert 173 at layer 25, with $\Delta W = 0.124$ (Table 1). Both components of the weight rise on redflag generation: its selection rate climbs from 0.43 on the benign prompts to 0.81 on the redflag prompts, and its conditional weight from 0.16 to 0.24. Beneath it sits a broad supporting set, expert 45 at layers 19 and 31, expert 25 at layer 20, expert 216 at layers 15 and 27, expert 189 at layer 17, and others, each separating the two prompt classes but none rivalling the leader.

4.2 Aggregate prefill “leaders” are largely a padding artifact

Read naively, the prefill block appears led by expert 173 at layer 1 ($\Delta W = 0.068$). The per-token decomposition shows this is mostly the repeated filler piece: excluding it, the layer-1 figure falls

Layer	Expert	ΔW
<i>Finance-domain experts (finance – non-finance)</i>		
20	62	0.180
30	95	0.177
19	223	0.167
<i>Consequence/duty experts (consequence – neutral)</i>		
17	189	0.073
23	122	0.065
14	157	0.058
15	171	0.053

Table 2: Finance-versus-consequence control ($n = 12$ prompts, bucketed). Finance-domain experts (top) fire on finance including neutral finance and are not refusal experts; the broad consequence/duty experts (bottom) generalize across domains. Generation block. Buckets are intent-matched, not strictly token-paired.

to 0.0004, and the other top aggregate prefill rows (expert 222 at layer 2, expert 36 at layer 22) collapse similarly. The prompt-side candidates that survive the correction are expert 218 at layer 14 ($0.025 \rightarrow 0.022$), expert 173 at layer 25 ($0.037 \rightarrow 0.046$), and expert 45 at layer 19 ($0.030 \rightarrow 0.034$). We therefore rest the safety read on the generation block, and treat the headline prefill leader as a methodological warning: token-count padding can fabricate a prefill expert that a per-token decomposition dissolves.

4.3 The cluster tracks consequence and duty, not a topic

The finance-versus-consequence control separates two things a single redflag set would conflate (Table 2). Finance-domain experts, expert 62 at layer 20, expert 95 at layer 30, expert 223 at layer 19, and expert 214 at layer 16, fire strongly on finance prompts including neutral finance, with near-zero selection on non-finance, so they are topic experts, not refusal experts. The consequence-minus-neutral contrast, by contrast, is led by expert 189 at layer 17, expert 122 at layers 11 and 23, expert 157 at layer 14, expert 171 at layer 15, and expert 45 at layer 19, and it generalizes across medical, legal, physical-safety, inspection, and finance-consequence content. The safety-relevant cluster is better read as real-world-consequence and professional-duty routing than as a topic or a single expert.

4.4 Suppressing the leading carrier does not break refusal

At the strongest setting we tried, bias -2 , expert 173 is nearly gone from the generation routing: its redflag selection rate is 0.05, down from 0.81, and its A–B weight difference is 0.004, down from 0.124 (Table 3, Figure 1). All three redflag prompts still refuse. The collapse is graded across the four bias levels; the behavior is not.

The oxycodone prompt is refused at every level, and at -2 the refusal gets *more* explicit about misuse and overdose; the ayahuasca instruction is refused throughout; the 500% prompt is warned against throughout, its wording shifting from “impossible and dangerous” to “trap” at the strongest bias. The benign halves stay professional. The routed mass that leaves expert 173 reappears nearby: generation separation reallocates to experts 45, 185, 157, 189, 216, and 133. The strongest carrier switches off and the cluster carries the refusal.

Router bias on E173	ΔW (L25)	S^A	W^A
baseline	0.124	0.809	0.192
-0.25	0.093	0.664	0.141
-0.5	0.056	0.510	0.091
-1	0.041	0.394	0.056
-2	0.004	0.046	0.005

Table 3: Dose-response of suppressing expert 173, read at its strongest carrier (layer 25, generation block). Routed mass and selection rate collapse toward zero. At every level the model still refuses or warns on all three redflag prompts; see text.

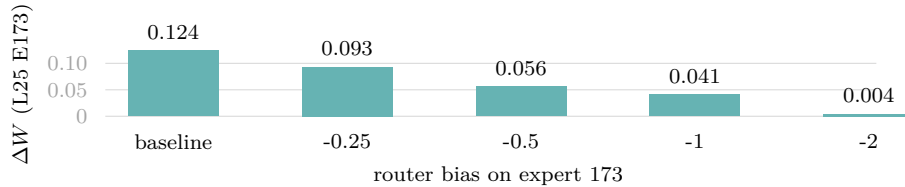


Figure 1: Generation-block A–B routing difference for expert 173 at layer 25 as its router bias is swept from baseline to -2 . The carrier collapses dose-dependently; the model’s refusals do not.

5 Interpretation

Three results point the same way, and only the last is a real test. One expert leads the routing, but it leads a cluster; the cluster tracks consequence rather than topic; and removing the leader does not stop the refusing. The suppression is the load-bearing one, because it is causal. A behavior owned by a single gate should weaken when the gate is shut. This one does not. In this model, harm-refusal is a property of a cluster that survives the loss of its strongest member.

This is the expert-routing analog of the distributed-representation expectation (Elhage et al., 2022; Bricken et al., 2023), and it stands against a strict single-locus reading of refusal carried over from dense models (Arditi et al., 2024). The contrast with Lai et al. (2025), which locates vulnerability in identifiable safety-critical experts, we take to be setup-dependent rather than a contradiction: a single-expert intervention that fails to bypass safety here may succeed in another model, prompt regime, or with a multi-expert intervention. What this run shows is the negative, that the strongest carrier in this model is not a sufficient point of attack on this set.

We are careful about what the suppression does and does not establish. It tests the *necessity* of expert 173 alone, and finds it not necessary; it does not test the joint necessity of the cluster, which a combined suppression of the replacement set (experts 45, 189, 157, 122) would address. The behavioral read is a direct reading of generated text on a small set, not a benchmarked refusal rate. The result is a structural observation about where the routing lives, offered with the controls that would harden it.

6 Limitations and falsification tests

What the present evidence does not settle.

- **Six and twelve prompts, one trajectory each.** Single greedy generations, so these are

point estimates with no sampled spread and no significance test, and per-prompt deltas can hang on tie-breaking. The “still refuses” verdict is the author’s reading of the text, not a graded harmful-completion rate.

- **We suppressed one expert.** Whether the cluster jointly carries the refusal is untested, so “no jailbreak” means removing the strongest carrier, not removing the cluster.
- **The control is bucketed, not token-paired.** Its finance-versus-consequence separations are weaker evidence than the matched pairs.

Tests that would confirm or break the claim.

- Jointly suppress the replacement set (experts 45, 189, 157, 122) in small steps and ask whether refusal survives the removal of the cluster, not just its leader; the distributed reading predicts graceful degradation, a gate reading predicts a cliff.
- Score behavior against a standardized red-team suite (Mazeika et al., 2024) under suppression, replacing the direct reading with a harmful-completion rate.
- Repeat with sampled decoding and more prompts per condition to attach confidence to the per-expert deltas and the behavioral counts.
- Run the same probe on other MoE families and on refusal-reduced fine-tunes to test whether the distributed pattern, or a localizable safety expert (Lai et al., 2025), is the general case.
- Compare the router-level picture to the residual-direction picture (Arditi et al., 2024) on the same model, to ask whether a single refusal direction coexists with distributed expert routing.

7 Conclusion

In base QWEN3.5-35B-A3B, harm-refusal does not route through a single safety expert. One expert leads, but biasing it to near-zero routed mass leaves the refusals intact, because the surrounding consequence-and-duty cluster takes the load. For anyone hoping to locate, monitor, or attack refusal at the level of a single MoE expert, the lesson here is that the obvious target is not the gate it looks like. The evidence is a small case study; the cluster-suppression and benchmark controls that would carry it past one model are set out in Section 6.

Reproducibility and Data and Code Availability

The numbers in this paper are transcribed from the analysis tables of a single capture session and are listed claim-by-claim, with source files, in the accompanying `SOURCES.md`. The supporting data accompany the paper in the `data/` directory: the run’s results write-up and journal, the all-layer A–B delta tables, the per-token decomposition that backs the filler-artifact correction, the finance-versus-consequence bucket tables, the expert-173 dose-response tables, the generated openings under suppression, and the prompt sets. Raw router tensors (`.npy`) were not retained in version control by project policy; the analysis tables above are derived from them and substantiate the reported statistics. The routing reconstruction satisfies $W = S \cdot Q$ per token set; we report the reconstructed, log-derived routing tables. No new model training was performed.

Ethical Considerations

This work studies how a model refuses, using a small set of synthetic redflag prompts (drug misuse, a prompt-injection attempt, and a financial “sure-thing”) that the model refused at every intervention level. We release the prompts and the model’s refusals, not harmful content: no completion in this study provides operational harm, and the suppression result is reported because it *fails* to bypass safety, not because it succeeds. The purpose is defensive, to understand whether refusal in an MoE router is concentrated enough to be a single point of failure for auditing or attack; our finding that it is distributed argues against a single-expert attack surface in this model, while we are explicit that this does not generalize without the tests in Section 6.

AI Collaboration Statement

Generative AI (Anthropic’s Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All experiments, captures, and numerical results are the author’s; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record to exist and to support the statement it is attached to. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content, including any errors.

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This work received no external funding. The author declares no competing interests. The model studied is a third-party open-weights release used as-is for measurement; the author has no affiliation with its creators.

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